Fares Mohamed

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Robotics, AI & Intelligent Systems Engineer

Professional Summary

Computer Science graduate ranked first in class with hands-on experience building ESP32 robots, vision-guided autonomous behavior, robot control interfaces, and real-time computer-vision systems. Combines Python, C++, embedded systems, AI/ML, and robotics engineering with professional experience teaching robotics, AI, programming, electronics, and project-based technical workshops.

Core Strengths

Robotics & Embedded: ESP32, Arduino, sensors, circuits, servo calibration, pose sequencing, controller mapping, FK/IK-assisted motion
AI & Vision: Python, OpenCV, MediaPipe, Ultralytics YOLO, object detection, live camera pipelines, ML, neural networks, CNNs
Instruction & Mentoring: Robotics/AI instruction, curriculum development, project-based learning, debugging, technical demonstrations, workshops

Relevant Experience

Engineering for Kids-Egypt 2023–2025
AI & Robotics Instructor (Seasonal)

- Taught students aged 11–16 robotics, AI, Arduino, circuits, Python, and C++ through hands-on project-based programs.
- Guided hardware/software builds from circuit assembly and sensor integration through programming, testing, debugging, and demonstrations.

Armstrong 2023–2025
AI & Automation Engineer / Technical Content Creator (Part-time)

- Developed robotics, AI, Arduino, circuits, and programming curricula, instructional media, and classroom-ready technical assets.
- Built AI-assisted content pipelines using n8n, custom GPTs, open-source models, scripts, and visual-generation tools for the Digital Egypt Cubs Initiative.

Kayfa Aug–Dec 2025
AI Subject Matter Expert

- Authored implementation-focused ML lessons, assignments, and Python notebooks covering neural networks, CNNs, clustering, anomaly detection, and unsupervised learning.

Selected Robotics & AI Projects

Biped Robot Control System
Python, ESP32, Computer Vision, Streamlit

- Built a six-servo biped with calibration, flash-backed pose storage, network discovery, live Android-camera input, robot/ball tracking, and autonomous forward motion that stops at a vision-based collision threshold.

DUM-E – ESP32 Robotic Arm Platform
Python, C++, ESP32, FK/IK, Streamlit

- Built firmware and operator tooling for per-joint calibration, controller mapping, sequence recording/replay, FK/IK-assisted planning, and guarded pick-and-place execution.

VisionDeck – Computer Vision Suite
Python, OpenCV, MediaPipe, YOLO

- Built real-time face/landmark, hand-gesture, and object-detection workflows with webcam and Android IP-camera support; included YOLOv8 fine-tuning on a custom 3-class subset.

SAGA – Agentic Narrative Intelligence Platform
Python, NLP, LangGraph, FastAPI, PostgreSQL

- Building an evidence-grounded AI analysis platform using structured extraction, evaluation workflows, local inference, durable jobs, and modern LLM orchestration patterns.

Education

Alexandria University, Faculty of Computer and Data Science 2022–2026
B.Sc. Computer Science (Intelligent Systems) | **Class Rank: 1st**

Additional Skills, Leadership & Certifications

Software: FastAPI, Streamlit, React/Next.js, TypeScript, SQL, PostgreSQL, Docker, Git, REST APIs | **Languages:** Arabic, English
Leadership: Head of AI Committee, ElCoder – led AI/ML workshops.
Certifications: Machine Learning Specialization; Deep Learning Specialization; Google Cloud Big Data & Machine Learning Fundamentals